NYC Airbnb Data Exploration Report

Executive Summary

This report presents an in-depth exploration of the 2019 NYC Airbnb dataset containing 48,895 listings across the five boroughs. Through statistical analysis and visualization, we've identified key patterns regarding property distribution, pricing dynamics, and host behaviors that inform our understanding of the NYC short-term rental market. Our exploration reveals significant geographic pricing disparities, room type preferences that vary by borough, and strategic minimum night requirements that affect pricing and occupancy. These insights provide a foundation for developing predictive models and extracting actionable business recommendations for market participants.

1. Geographic Distribution and Room Type Analysis

1.1 Room Type Distribution by Borough

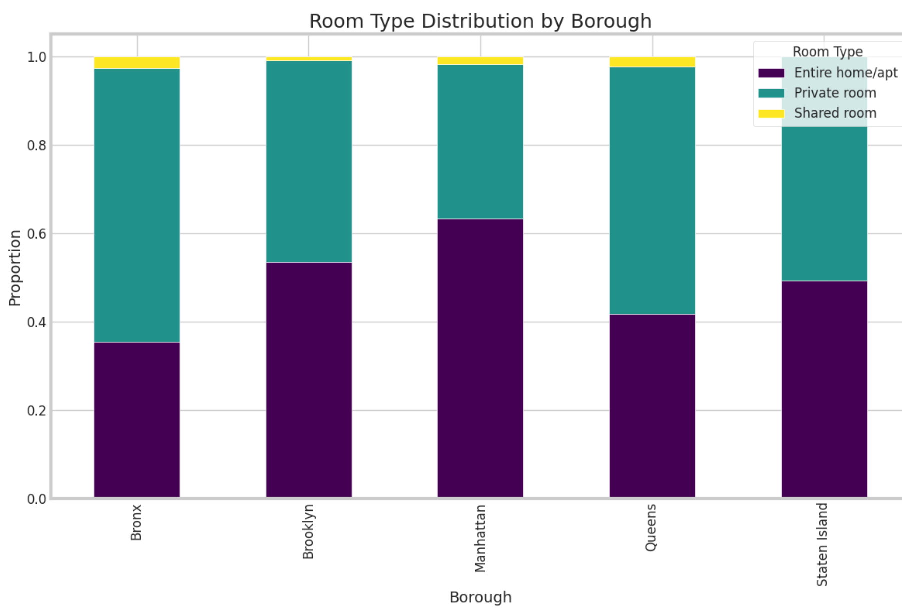


Figure 1: Room Type Distribution by Borough

The visualization of room type distribution across boroughs reveals several notable patterns:

- **Manhattan** shows the highest proportion of entire homes/apartments (63%), indicating its position as a premium market for tourists seeking private accommodations.

- **The Bronx** has the lowest proportion of entire homes/apartments (36%) and the highest share of private rooms (60%), suggesting a more budget-oriented or residential character.
- **Brooklyn** maintains a balanced distribution with 53% entire homes, 45% private rooms, and 2% shared rooms.
- **Shared rooms** constitute a minimal percentage across all boroughs (1-3%), indicating limited market demand for this accommodation type.

This distribution aligns with the general real estate market in NYC, where Manhattan properties command premium prices and outer boroughs offer more affordable options. The higher proportion of entire homes in Manhattan suggests that the return on investment for dedicating entire properties to short-term rental is more favorable in high-demand tourist areas.

1.2 Availability Distribution by NYC Borough

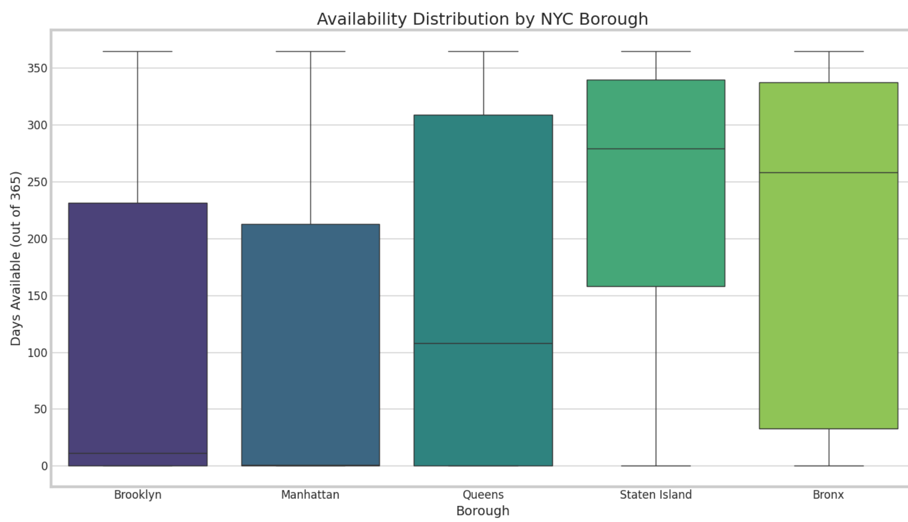


Figure 2: Availability Distribution by NYC Borough

The boxplot of availability patterns reveals interesting insights about market dynamics:

- **Staten Island and the Bronx** show the highest median availability (approximately 275 days per year), suggesting lower occupancy rates or less consistent demand.
- **Manhattan and Brooklyn** display the lowest median availability (approximately 15-20 days), indicating high demand and better occupancy rates.
- **Queens** occupies a middle position with median availability of about 110 days.

- **All boroughs** show significant variability in availability (wide interquartile ranges), highlighting the diversity of hosting strategies.

These patterns suggest that Manhattan and Brooklyn properties operate in a seller's market where hosts can afford to be selective with availability, potentially limiting days available to maintain premium pricing. Conversely, Staten Island and Bronx listings may need to offer greater availability to achieve desired occupancy rates.

2. Pricing Dynamics and Patterns

2.1 Average Price by Borough and Room Type

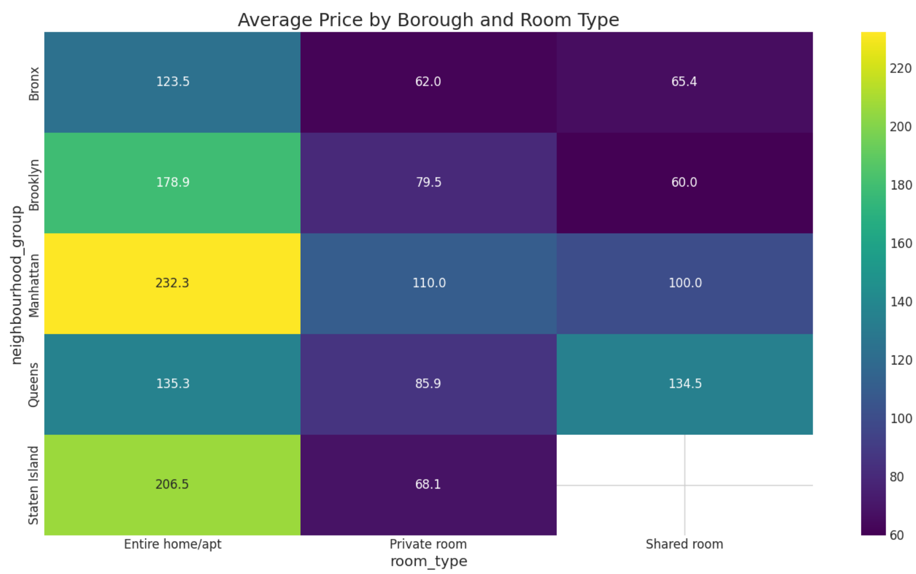


Figure 3: Average Price by Borough and Room Type

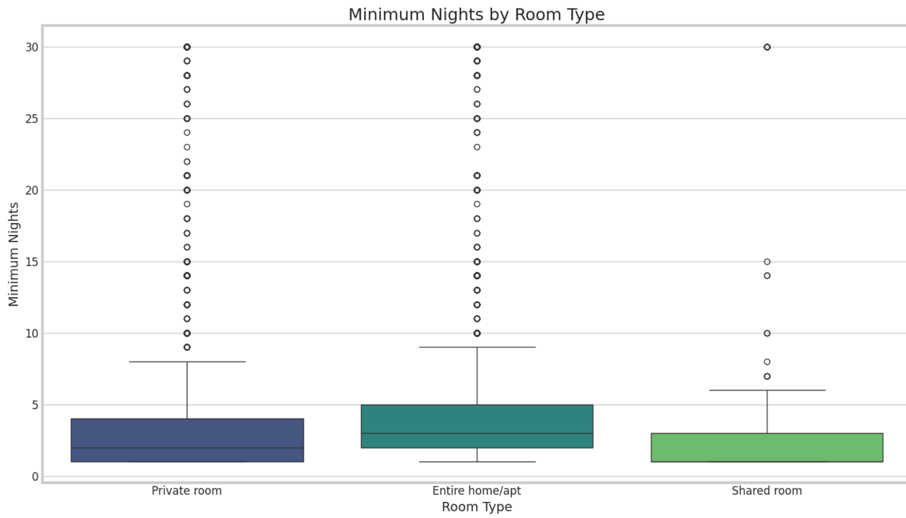
The heatmap reveals clear pricing patterns across boroughs and room types:

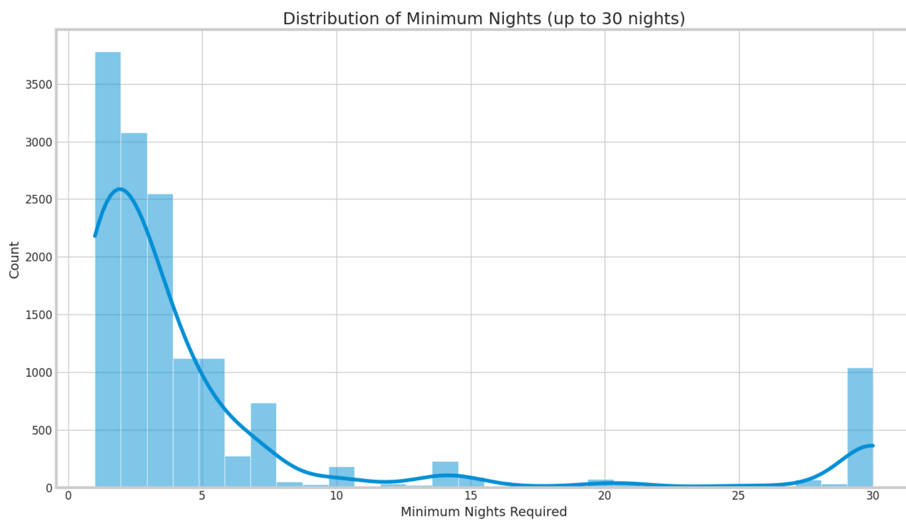
- **Manhattan** commands the highest prices across all room types, with entire homes averaging $232.30 per night—significantly higher than other boroughs.
- **Staten Island** shows surprisingly high prices for entire homes ($206.50), possibly due to larger property sizes or limited supply.
- **Brooklyn** follows as the third most expensive borough for entire homes ($178.90).
- **The Bronx** offers the most affordable entire homes ($123.50).
- **Shared rooms** in Queens show an anomalously high average price ($134.50), warranting further investigation.

- **Private rooms** show less price variation across boroughs than entire homes, ranging from $62.00 (Bronx) to $110.00 (Manhattan).

The substantial price premium for entire homes versus private rooms (approximately 2-3x) demonstrates the significant impact of room type on pricing strategy. The price gradient from Manhattan to outer boroughs reflects established neighborhood value perceptions and tourist demand patterns.

2.2 Minimum Nights Requirements





The analysis of minimum night requirements reveals strategic host behaviors:

- **Entire homes/apartments** have slightly higher median minimum night requirements (3 nights) compared to private rooms and shared rooms (2 nights).

- **All room types** show numerous outliers with requirements of 10+ nights, suggesting some hosts target longer-term stays.
- **The distribution** shows strong peaks at common values:
  - 1-night minimums (most common): Approximately 3,800 listings
  - 2-night minimums: Approximately 3,100 listings
  - 3-night minimums: Approximately 2,500 listings
  - 30-night minimums (notable spike): Approximately 1,000 listings

The 30-night spike likely represents hosts complying with regulations in certain areas that prohibit rentals of less than 30 days or hosts targeting extended business travelers or temporary relocations. The overall distribution shows that most hosts prefer shorter minimum stays (1-3 nights), optimizing for the typical weekend or short vacation traveler.

3. Guest Experience and Host Behavior Patterns

3.1 Average Number of Reviews by Room Type

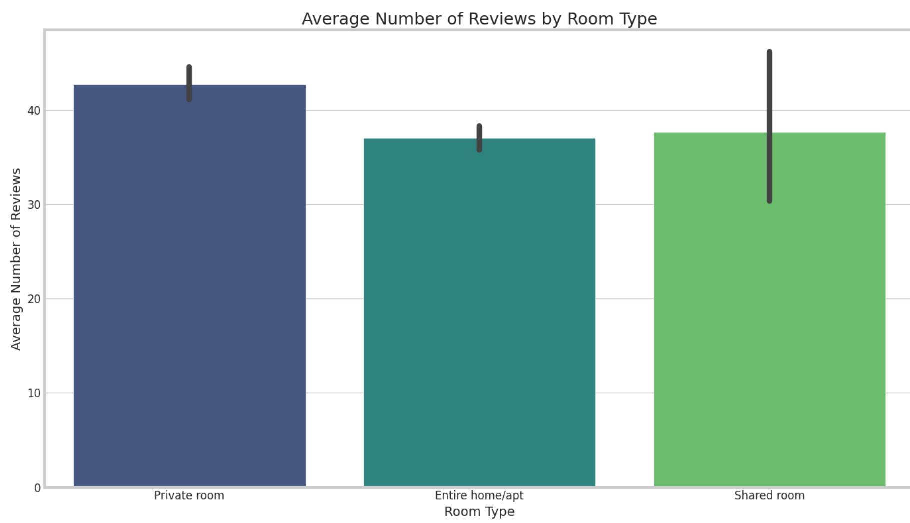


Figure 4: Average Number of Reviews by Room Type

The review count analysis provides insights into guest preferences and host experience:

- **Private rooms** accumulate the highest average number of reviews (approximately 42), suggesting higher turnover rates or potentially longer listing history.
- **Shared rooms and entire homes** show similar average review counts (approximately 37).

- **All room types** display significant variance in review counts (shown by error bars), indicating wide differences in host experience or property popularity.

The higher review count for private rooms might indicate that this accommodation type attracts more frequent, shorter stays, generating more reviews per time period. It could also suggest that hosts of private rooms may be more active in soliciting reviews than other hosts.

4. Key Relationships and Correlations

4.1 Price and Availability Relationship

From the visualizations, we can infer an inverse relationship between price and availability:

- High-priced areas (Manhattan, Brooklyn) show lower availability days.
- More affordable areas (Bronx, Staten Island) show higher availability days.

This suggests a supply-demand dynamic where hosts in premium locations can maintain high occupancy despite higher prices, while hosts in less desirable locations may need to offer greater availability to achieve satisfactory booking rates.

4.2 Room Type and Review Patterns

The data reveals interesting correlations between room type and guest engagement:

- Private rooms receive more reviews despite generally lower prices.
- Entire homes command higher prices but don't necessarily generate more reviews.

This suggests different guest demographics or use cases: private rooms may attract more frequent, budget-conscious travelers, while entire homes may appeal to families or groups who stay less frequently but pay premium prices.

5. Market Segmentation Insights

Based on the exploratory analysis, we can identify several distinct market segments:

5.1 Premium Urban Experience (Manhattan Entire Homes)

- Highest prices ($232.30 average)
- Limited availability (low median days available)
- Catering to tourists seeking complete privacy and premium locations

5.2 Value Accommodations (Brooklyn and Queens Private Rooms)

- Mid-range prices ($79.50-$85.90)
- Higher review counts
- Attractive to budget-conscious travelers seeking authentic local experiences

5.3 Extended Stay Market (30+ Minimum Night Listings)

- Significant subset across all boroughs
- Likely targeting business travelers, students, or temporary relocations
- Different pricing and availability strategies than short-term rentals

5.4 Outer Borough Value Properties (Bronx and Staten Island)

- More affordable prices
- Higher availability
- Potential investment opportunities with lower entry costs

6. Unexpected Insights and Anomalies

6.1 Staten Island Pricing

Staten Island shows surprisingly high prices for entire homes ($206.50), second only to Manhattan. This contradicts typical real estate value perceptions and warrants further investigation. Possible explanations include larger property sizes, waterfront locations, or limited supply increasing prices.

6.2 Queens Shared Room Pricing

Shared rooms in Queens show unusually high average prices ($134.50), comparable to entire homes. This anomaly could result from a small sample size, unique property characteristics, or specific neighborhood effects.

6.3 Bimodal Minimum Night Distribution

The strong spike in 30-night minimum requirements suggests a regulatory or strategic segmentation in the market between short-term and medium-term rentals. This bimodal distribution may necessitate separate modeling approaches for these distinct market segments.

7. Hypotheses for Further Investigation

Based on our exploration, we propose the following hypotheses for further investigation:

7.1 Geographic Pricing Hypothesis

H1: Neighborhood-level factors (proximity to tourist attractions, public transportation, safety) are stronger predictors of price than borough-level designation.

7.2 Review Impact Hypothesis

H2: Higher review counts correlate with pricing optimization, with experienced hosts (more reviews) showing more market-aligned pricing strategies.

7.3 Minimum Night Strategy Hypothesis

H3: Hosts strategically set minimum night requirements based on location desirability, with prime locations having lower minimum requirements due to consistent demand.

7.4 Seasonal Availability Hypothesis

H4: Availability patterns show seasonal variations, with hosts in premium areas restricting availability during peak tourist seasons to maximize revenue.

8. Recommendations for Modeling Approach

Based on our exploratory findings, we recommend the following modeling approaches:

8.1 Segmented Modeling

Develop separate predictive models for different market segments: - Short-term rentals (1-7 night minimums) - Medium-term rentals (8-29 night minimums) - Long-term rentals (30+ night minimums)

8.2 Geographic Hierarchical Features

Incorporate nested geographic features to capture location effects: - Borough-level factors - Neighborhood-level factors - Proximity to key attractions/transit

8.3 Interaction Terms

Include interaction terms in models to capture relationships between: - Room type and borough (pricing varies differently by room type across boroughs) - Minimum nights and borough (different optimal strategies by location) - Availability and room type (different patterns by accommodation type)

8.4 Feature Engineering Priorities

Focus on creating derived features that capture: - Revenue potential (price × availability) - Competitive positioning (price relative to neighborhood average) - Listing experience (review count as a proxy for host experience) - Occupancy estimation (availability patterns)

9. Conclusion

The NYC Airbnb dataset reveals a complex market with significant geographic stratification, distinct hosting strategies, and clear price differentials across room types and boroughs. Manhattan and Brooklyn emerge as premium markets with higher prices and lower availability, while outer boroughs offer more affordable options with higher availability.

Room type significantly impacts pricing across all boroughs, with entire homes commanding substantial premiums over private rooms. Minimum night requirements reveal strategic host behaviors, with a notable segment targeting extended stays (30+ nights).

These insights provide a foundation for developing sophisticated predictive models and extracting actionable business recommendations for hosts and investors in the NYC short-term rental marketplace. The identified market segments, anomalies, and hypotheses will guide our modeling approach and feature engineering efforts in subsequent phases of the project.